

Karan Kumar

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PROFESSIONAL SUMMARY

Full Stack Developer (Intern) with hands-on experience building production-style web applications, with growing exposure to AI-driven features. Strong at translating requirements into working features, integrating APIs, and building interactive UIs. Actively strengthening data structures, backend fundamentals, and system thinking to perform effectively in technical interviews and real-world development environments.

WORK EXPERIENCE

Web Developer Intern – Central Coalfields Limited, CCL Jul 2025 – Aug 2025 | Ranchi

- Developed a prototype decentralized e-commerce platform as part of an internal innovation initiative, combining traditional web architecture with Web3 components.
- Implemented smart contract-based transaction logic to simulate transparent order tracking and immutable records.
- Built responsive UI components using React and Tailwind CSS, focusing on usability and performance across devices.
- Collaborated with mentors to integrate frontend and backend workflows, improving understanding of real-world development cycles and deployment constraints.

Software Engineering – JP Morgan Chase & Co. Dec 2026 – Jan 2026 | Virtual

- Built a Spring Boot microservice to process high-volume financial transactions using Apache Kafka, simulating real-world asynchronous event-driven systems.
- Implemented transaction validation and persistence using Spring Data JPA with an H2 SQL database, modeling relational user and transaction entities.
- Integrated an external REST-based Incentive API, applying business rules to recipient balances while maintaining transactional correctness.
- Developed and tested REST API endpoints using Spring MVC, validating system behavior with JUnit, embedded Kafka, and IDE debugging tools.

EDUCATION

Amity University Jharkhand – Ranchi Jul 2027

Bachelor of Technology, Computer Science and Engineering (B. Tech. CSE)

CGPA: 7.5 (up to 6th Semester)

Kendriya Vidyalaya Hinoo – Ranchi Jul 2022

12th CGPA: 7.68 | 10th CGPA: 9.16

VOLUNTEERING & LEADERSHIP

Player – District Level Cricketer Feb 2019 – Sep 2023 | Ranchi

- Represented Ranchi in district tournaments; honed leadership, teamwork, and strategic planning skills under high-pressure competitive environments.

Event Coordinator Apr 2024 – Apr 2024 | Amity

- Organized technical workshops and coding events, managing logistics and fostering a collaborative coding culture with sports events on campus.

CERTIFICATIONS

AI Practitioner & Data Analytics – Infosys Springboard Jun 2025

Cyber Security & Technology Consulting – Deloitte Dec 2025

Database Fundamentals – Infosys Springboard Sep 2023

Generative AI & Prompt Engineering – Infosys Springboard Jun 2025

SKILLS

Languages: Java, JavaScript (ES6+), SQL
Frontend Development: HTML, CSS, React.js
Backend & APIs: Node.js, REST APIs, Spring Boot (basic)
Databases: PostgreSQL (Drizzle ORM), MongoDB
AI & Intelligent Features: Google Gemini API, Prompt Engineering, Genetic Algorithms (applied in simulations)
Tools & Platforms: Git, GitHub, VS Code, Vite

PROJECTS

Glimmer — Cosmic Social Platform

Aug 2025 – Dec 2025

- Built a full-stack social web application using React, Node.js, PostgreSQL, and WebSockets, supporting real-time messaging and interactive user experiences.
- Implemented real-time chat with ephemeral messages (auto-expiry after 24 hours), reducing storage usage and improving privacy.
- Designed a 3D visualization module using Three.js, converting user content into a navigable “memory universe.”
- Integrated AI-powered interactions using Google Gemini API to generate contextual responses and dynamic in-app behavior.
- Tech Stack: React, Node.js, WebSockets, PostgreSQL, Three.js, Gemini API

reuselT Electronics

Jul 2025 – Sep 2025

- Built a prototype for buying and selling refurbished electronics, integrating Ethers.js for secure wallet connection (MetaMask) and crypto-based authentication.
- Implemented an AI Shopping Assistant using Google Gemini API for smart cart summaries and a DIY Studio interface allowing users to design custom device skins.
- Designed an Auction House for bidding on rare collectibles and a Milestone system rewarding user engagement with NFT-based badges.
- Tech Stack: React, Node.js, Ethers.js, Web3.js, Gemini API

AI-Powered Train Traffic Control

Sep 2025 – Sep 2025

- Created an AI-based simulation system to analyze and resolve train scheduling conflicts using genetic algorithms.
- Compared manual scheduling logic vs AI-assisted optimization to demonstrate throughput and safety improvements.
- Developed backend simulation logic using Spring Boot, focusing on algorithm correctness rather than UI complexity.
- Tech Stack: Java, Spring Boot, Genetic Algorithms